

DWIJESH DOOKRAZ

SOFTWARE ENGINEER | BACKEND & MACHINE LEARNING SYSTEMS

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PROFESSIONAL SUMMARY

Performance-driven Software Engineer with proven expertise building scalable backend architectures, high-throughput APIs, and applied machine learning pipelines. Adept at transforming complex algorithmic models into production-grade systems using Python, FastAPI, and cloud frameworks. Strong background designing low-latency cloud workflows, data pipelines, and responsive full-stack interfaces.

TECHNICAL SKILLS

- Programming Languages:** Python, TypeScript, JavaScript, SQL
- Frameworks & Libraries:** FastAPI, Flask, React, Next.js 15, PyTorch, scikit-learn, OpenCV, NumPy, Optuna, Monaco Editor, Playwright
- Cloud & Infrastructure:** Google Cloud Platform (GCP), GCP Cloud Run, Firestore, Sentry, Git, CI/CD pipelines
- Protocols & Architectures:** RESTful APIs, OAuth2, Webhooks, Microservices, Multi-channel Systems

PROFESSIONAL EXPERIENCE

- Nusmark — Backend Engineer** 2024 – 2026
- Engineered a resilient bidirectional sync system for Google Calendar and Outlook using Python, FastAPI, and OAuth2, eliminating duplicate event creation via transactional Firestore writes and webhook lifecycle management.
 - Architected and deployed a multi-modal WhatsApp AI Assistant using Python and Flask on GCP Cloud Run, parsing text, audio, and image intent via custom NLP models to automate calendar onboarding and event generation.
 - Designed an enterprise-grade multi-channel notification engine using Firestore and Microsoft Graph APIs, orchestrating reliable real-time push reminders and lifecycle synchronization across cross-provider ecosystems.

TECHNICAL PROJECTS

- rPPG Heart Rate Estimation — Final-Year Dissertation** 2024 – 2025
- Developed a DeepPhys dual-stream attention CNN using PyTorch and OpenCV for sensorless heart rate extraction from facial videos, achieving Pearson $r = 0.87$ and MAE 8.85 BPM on the UBFC-rPPG benchmark.
 - Mitigated demographic bias across cross-population data via localized Python and NumPy pipelines, successfully reversing negative baseline correlations from $r = -0.23$ to $r = +0.59$.
 - Optimized hyperparameters using an Optuna Bayesian framework backed by subject-aware k-fold cross-validation, outperforming traditional ICA, PCA, and frequency band-pass baselines.
- VS Code Portfolio — dwijesh.dev** 2026
- Built a responsive browser development shell using Next.js 15, TypeScript, and React, securely routing live repository insights via server-side Next.js route handlers and the official GitHub API.
 - Enforced code quality pipelines by implementing Sentry runtime error monitoring, Vercel analytics tracking, and automated Playwright UI end-to-end testing suites for stable production deployment.
 - Structured a backend proxy engine to securely fetch and dynamically inject visual badge streams into the Monaco Editor workspace, successfully bypassing browser Content Security Policy (CSP) restrictions.

EDUCATION

- University of Southampton** 2022 – 2025
- BSc Computer Science — First Class Honours**
- Key Academic Highlights: Final Dissertation (82%), Social Computing Techniques (83%), Machine Learning Technologies (80%), Cloud Application Development (79%), Software Engineering Group Project (73%).*